



# Assignment

1. Chebyshev polynomials are defined recursively. Chebyshev polynomials are separated into two kinds: first and second. Chebyshev polynomials of the first kind,  $T_n(x)$ , and of the second kind,  $U_n(x)$ , are defined by the following recurrence relations:

$$T_n(x) = \begin{cases} 1 & \text{if } n = 0 \\ x & \text{if } n = 1 \\ 2xT_{n-1}(x) - T_{n-2}(x) & \text{otherwise} \end{cases} \quad (3)$$

$$U_n(x) = \begin{cases} 1 & \text{if } n = 0 \\ 2x & \text{if } n = 1 \\ 2xU_{n-1}(x) - U_{n-2}(x) & \text{otherwise} \end{cases} \quad (4)$$

Write a function `my_chebyshev_poly1(n,x)`, where the output `y` is the  $n$ -th Chebyshev polynomial of the first kind evaluated at  $x$ . Be sure your function can take list inputs for  $x$ . You may assume that  $x$  is a list. The output variable, `y`, must be a list also.



# Assignment

- What to submit:
- Take a screenshot that covers your code and the output from the following

```
def my_chebyshev_poly1(n,x):  
    # Write your function code here  
  
    return y
```

```
x = [1, 2, 3, 4, 5]
```

```
# Output: [1, 1, 1, 1, 1]  
my_chebyshev_poly1(0,x)
```

```
# Output: [1, 2, 3, 4, 5]  
my_chebyshev_poly1(1,x)
```

```
# Output: [1, 26, 99, 244, 485]  
my_chebyshev_poly1(3,x)
```